

SVKM'S NMIMS
MUKESH PATEL SCHOOL OF TECHNOLOGY MANAGEMENT & ENGINEERING
SCHOOL OF TECHNOLOGY MANAGEMENT

Academic Year: 2024-2025

Programs: B Tech / MBA Tech / BTI

Year: III/V Semester: V/IX

Stream: All Applicable Streams / MECHA / COM

Subject: Elements of Project Management

Time: 03 hrs (2:00pm to 5:00pm)

Date: 11/12/24

No. of Pages: 03

Marks: 100

Final Examination (2024-25) / Re Exam 23-24

Instructions: Candidates should read carefully the instructions printed on the question paper and on the cover of the Answer Book, which is provided for their use.

- 1) Question No. 1 is compulsory.
- 2) Out of remaining questions, attempt any 4 questions.
- 3) **In all 5 questions to be attempted.**
- 4) All questions carry equal marks.
- 5) **Answer to each new question to be started on a fresh page.**
- 6) **Figures in brackets on the right hand side indicate full marks.**
- 7) **Assume Suitable data if necessary.**

- Q1** Answer briefly:
- | | | |
|------------------------|--|-----|
| CO-1;
SO-1;
BL1 | a. State the practical importance of Project management in industries such as construction, manufacturing, and IT. | [5] |
| CO-1;
SO-4;
BL-2 | b. "The project manager is the centerpiece of project management" Explain the statement. | [5] |
| CO-2;
SO-4;
BL-2 | c. List and explain benefits of Responsibility Assignment Matrix | [5] |
| CO-3;
SO-1;
BL-2 | d. Discuss the importance of information monitoring and control in project risk management. | [5] |

- Q2**
- | | | |
|------------------------|---|------|
| CO-1;
SO-7;
BL-4 | a. 'Scope' and 'Feasibility' are two important and significant concepts in project management. Explain the statement in detail. | [10] |
|------------------------|---|------|

- CO-1;
SO-7;
BL-2
- b. Describe the role of organization structures in project management and list the different organization structures. [10]

Q3

- CO-2;
SO-1;
BL-2
- a. List and explain steps for construction of WBS along with an example. [10]
- CO-3;
SO-1;
BL-2
- b. Explain the steps involved in the cost estimating process for a project. How can these estimates be used in project management? [10]

Q4

- CO-3;
SO-1;
BL-2
- a. Explain the concept of risk analysis in Project Management. Provide a detailed overview of the methods and techniques used for risk analysis, including qualitative and quantitative approaches. [10]
- CO-3;
SO-1;
BL-2
- b. Describe the key deliverables, such as the final project report and post-implementation review and discuss the challenges that may arise during the project handover. [10]

Q5

- CO-2;
SO-7;
BL-1
- a. State the role that Gantt charts play in the planning and scheduling process, and also state their advantages and disadvantages. [10]
- CO-3;
SO-2;
BL-3
- b. Illustrate the challenges encountered in managing the budget, including any unforeseen expenses or scope changes. State the techniques were used to ensure the project stayed within budget. [10]

Q6

- CO-1;
SO-1;
BL-1
- a. Define Project management in terms of triple constraints. Provide examples of challenges related to balancing these triple constraints and explain the strategies used to address them with an example. [10]
- CO-3;
SO-1;
BL-1
- b. Explain the process of monitoring and control in project risk management. How does it impact project performance and cost control? [10]

Q7

CO-1;
SO-1;
BL-2

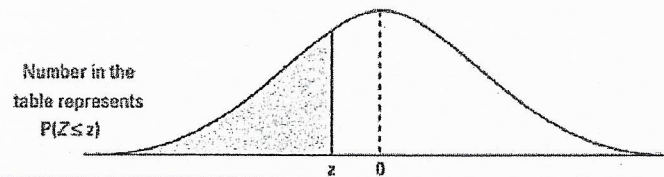
- a. Explain the key types of feasibility and how they contribute to informed decision-making. Provide examples of challenges faced during feasibility studies and strategies to address them. [8]

CO-2;
SO-2;
BL-3

- b. A project consists of the following activities, with their respective optimistic (O), pessimistic (P), and most likely (M) time estimates (in days): [12]

Activity	Precedence	Optimistic time (T_o)	Modal time (T_m)	Pessimistic time (T_p)
A	--	3	5	7
B	A	2	4	6
C	A	4	6	8
D	B	1	3	5
E	B	3	5	7
F	C	2	3	4
G	C	5	7	9
H	D, E, F, G	4	6	8
I	H	2	4	6

1. Calculate the expected duration (TE) for each activity.
 2. Determine the variance for each activity.
 3. Construct a project network diagram and identify the critical path, along with the total project duration.
 4. Calculate the probability that the project will be completed within 25 days.
- Refer normal distribution table:



z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
-2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
-2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
-2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
-2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
-2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
-2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
-2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
-1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233

